

Can a Quantum Circuit Read a Sentence?

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Encoding “the clever program runs” as a trained 2-qubit parameterized quantum circuit

Physics 14N · Stanford University

Motivation

We implement a **Quantum Natural Language Processing (QNLP)** pipeline that classifies sentence sentiment as **positive** or **negative** by executing a parameterized quantum circuit. Grammar structure determines the circuit topology; learned rotation angles encode word-level semantics. Classification follows from a single measurement in the computational basis.

$$\text{positive} \equiv |1\rangle \quad | \quad \text{negative} \equiv |0\rangle$$

End-to-End Architecture

words → **rotation parameters** · **grammar** → **circuit structure & weights** · **circuit** → **measure** → **class label**

Each word is represented by a single trainable rotation angle. Grammar determines how those angles are composed into a circuit. Measuring the output qubit yields $P(|1\rangle)$, interpreted as the probability of **positive** sentiment.

Step 1 — Grammatical roles and learned weights

All sentences share a common syntactic structure; each word is assigned a colored **grammatical role**:



Grammar determines the circuit topology and assigns role-specific weights to each word’s rotation parameter. Verbs and adjectives receive higher weights, reflecting their greater contribution to sentential polarity:

role	weight
verb	1.40
adjective	1.25
object-adjective	0.90
subject	0.45
object	0.35

(Object-role weights activate for sentences with an explicit object; our example, “the clever program runs,” uses only the top three.)

Step 2 — Qubit states and measurement

A classical bit takes a definite value in $\{0, 1\}$. A **qubit** is a two-level quantum system whose pure state is a unit vector on the **Bloch sphere**. Before measurement, the qubit occupies a **superposition** of computational basis states $|0\rangle$ and $|1\rangle$.

- An $R_y(\theta)$ rotation from $|0\rangle$ yields $|\psi\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$, a point at polar angle θ from the north pole of the Bloch sphere.
- Measurement** in the $\{|0\rangle, |1\rangle\}$ basis projects $|\psi\rangle$ onto one of the basis states. The probability of each outcome equals the **squared magnitude of the projection**: $P(|1\rangle) = |\langle 1|\psi\rangle|^2 = \sin^2(\theta/2)$. The equatorial state ($\theta = \pi/2$) gives $P(|0\rangle) = P(|1\rangle) = \frac{1}{2}$.

We use a **2-qubit register**: a **subject qubit** and a **sentence qubit** whose measurement yields the classification. The joint state lives in the tensor product space $\mathcal{H}_{\text{subj}} \otimes \mathcal{H}_{\text{sent}}$; amplitudes of product states multiply.

Step 3 — Encoding words as rotation angles

Each word is encoded as an $R_y(\theta)$ rotation angle applied to its qubit wire. Grammar composes the relevant angles onto each wire. For **the clever program runs**, the *trained* parameters are **clever**= 0.098, **program**= 0.043, **runs**= 0.489:

$$\text{subject } \theta = \text{clever} + \text{program} = 0.098 + 0.043 = \mathbf{0.141}$$

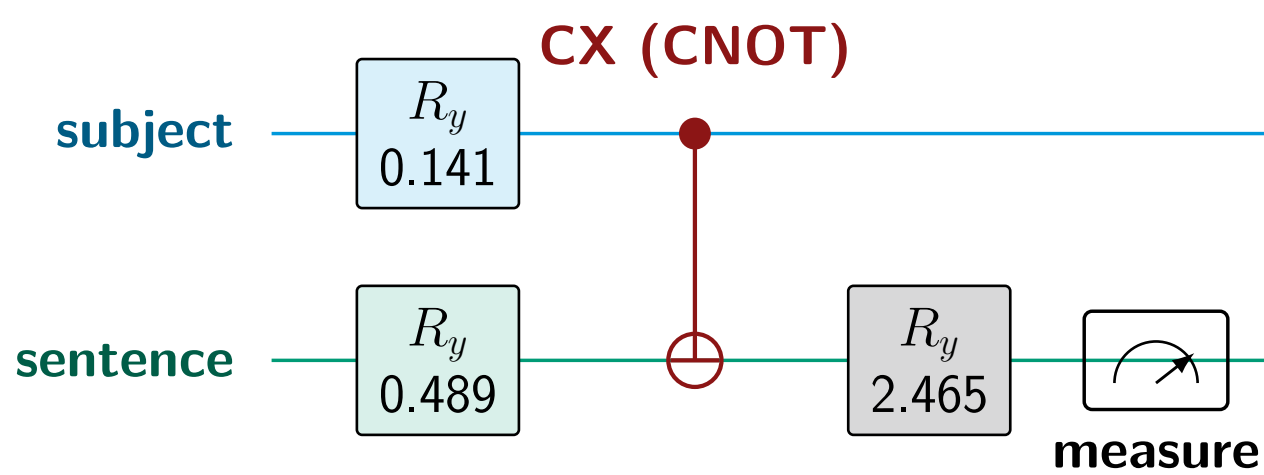
$$\text{sentence } \theta = \text{runs} = \mathbf{0.489}$$

$$\text{verdict angle} = \underbrace{\pi/2}_{\text{equatorial init.}} + \underbrace{\text{score}}_{\text{weighted sum}}$$

$$\text{score} = \underbrace{0.068}_{\text{bias}} + 1.25(0.098) + 0.45(0.043) + 1.40(0.489) = 0.894$$

$$\text{verdict angle} = \pi/2 + 0.894 = \mathbf{2.465}$$

The sentence qubit is initialized to the equatorial state ($\theta = \pi/2$), so the weighted score shifts its state vector toward either pole — the verb dominates (weight 1.40).



An $R_y(\theta)$ gate rotates the qubit state vector by angle θ about the y -axis of the Bloch sphere. The sentence qubit receives two successive R_y rotations: first by its word parameter (0.489), then by the verdict angle (2.465).

Step 4 — Entanglement via the CNOT Gate

Grammar requires the **subject** and **sentence** qubits to be treated as a single composite system. The **CNOT (CX) gate** encodes this grammatical composition:

If subject = $|1\rangle$: flip sentence qubit. If subject = $|0\rangle$: leave sentence qubit unchanged.

Action on the computational basis

The 2-qubit system has four computational basis states (left digit = subject, right = sentence). CNOT **swaps the two states where the subject is $|1\rangle$** ($|10\rangle \leftrightarrow |11\rangle$):

	00	01	10	11
before	0.968	0.241	0.068	0.017
after	0.968	0.241	0.017	0.068

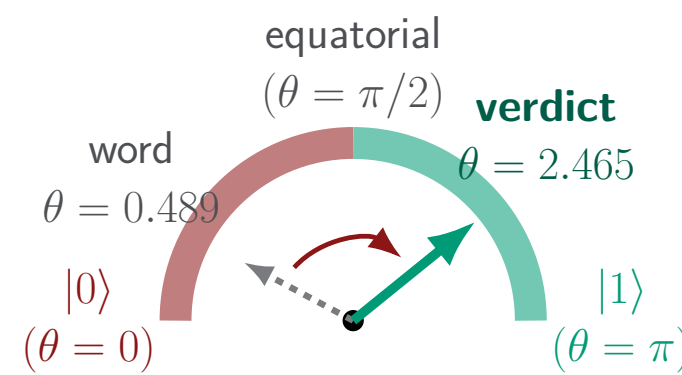
Because each qubit is in a **superposition**, the CNOT produces an **entangled** state: the post-gate state cannot be written as a tensor product $|\psi_{\text{subj}}\rangle \otimes |\psi_{\text{sent}}\rangle$ of individual qubit states.

Forward Pass: Full State Evolution

The 2-qubit state is tracked as four computational basis amplitudes ($|00\rangle, |01\rangle, |10\rangle, |11\rangle$; **left digit** = **subject**, **right** = **sentence**). Each gate reshuffles these amplitudes.

operation	00	01	10	11
initialized to $ 00\rangle$	1.000	0.000	0.000	0.000
apply word-parameter R_y to each qubit	0.968	0.241	0.068	0.017
CNOT: $ 10\rangle \leftrightarrow 11\rangle$	0.968	0.241	0.017	0.068
apply verdict $R_y(2.465)$ to sentence qubit	0.094	0.993	−0.059	0.039

The $R_y(\theta)$ rotation traces a path along the Bloch sphere: $\theta = 0$ leaves the qubit at $|0\rangle$ (north pole), $\theta = \pi$ rotates it to $|1\rangle$ (south pole), and $\theta = \pi/2$ yields the equatorial superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. The sentence qubit’s word angle ($\theta = 0.489$) places its state vector near the north pole; the verdict angle ($\theta = 2.465$) rotates it past the equator into the $|1\rangle$ hemisphere:



Measuring the sentence qubit in the $\{|0\rangle, |1\rangle\}$ basis: basis states $|01\rangle$ and $|11\rangle$ (right digit = 1, **green** columns) correspond to a positive outcome. The probability of each is the squared magnitude of its amplitude:

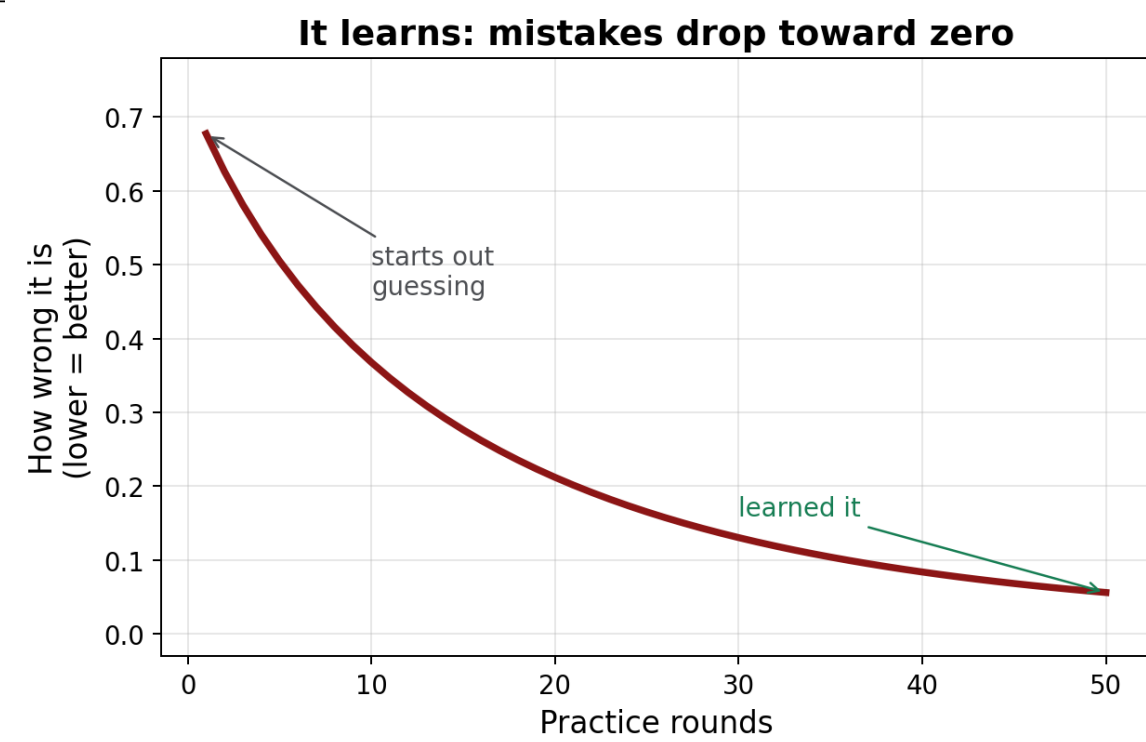
$$P(|1\rangle) = 0.993^2 + 0.039^2 = \mathbf{0.988} \Rightarrow \mathbf{POSITIVE} \checkmark$$

For **the broken code fails**, the same circuit gives $P(|1\rangle) = \mathbf{0.003} \Rightarrow \mathbf{NEGATIVE}$.

Classification Results

	Correct
Training (60)	100%
Test (12)	100%

All predictions are obtained by exact statevector simulation. With only **2 qubits**, simulation is exact and completes in < 1 s; full-scale QNLP would require many entangled qubits, motivating execution on actual quantum hardware.



Training: the parameter-shift rule

Evaluate the circuit to obtain $P(|1\rangle)$; the error is $e = P(|1\rangle) - y_{\text{label}}$. The gradient of $P(|1\rangle)$ with respect to each rotation angle θ is estimated via the *parameter-shift rule*: re-running the circuit with the angle shifted by $\pm \frac{\pi}{2}$ gives the exact gradient,

$$\frac{\partial P}{\partial \theta} = \frac{1}{2} [P(\theta + \frac{\pi}{2}) - P(\theta - \frac{\pi}{2})]$$
$$\theta \leftarrow \theta - \eta \left(e \cdot \frac{\partial P}{\partial \theta} \right)$$

This identity holds exactly for gates of the form $e^{-i\theta G/2}$, making it directly executable on quantum hardware. η is the learning rate; training proceeds for ~ 50 epochs.